

Q.1)

-1 π platform-Name ($\&$ Difficulty-Level = 'Advanced' (Course) $\bowtie$ ENROLLMENT $\bowtie$ PLATFORM)

Explanation:-

Selects course with difficulty level 'Advanced', join them with enrollment and platform table

-2 π Instructor-Name ($\&$ category = 'Data Science' $\wedge$ status = 'completed' (COURSE $\bowtie$ ENROLLMENT $\bowtie$ COURSE_ACTIVITY $\bowtie$ INSTRUCTOR))

Explanation:-

Finds instructor who handled Data science course where activity status is completed

-3 π platform-ID, platform-Name (Platform) - π platform-ID, platform-Name (PLATFORM $\bowtie$ INSTRUCTOR)

Explanation:-

Returns platform that do not appear in instructor table

-4 π course-ID, course-Name ($\&$ E1.Platform $\neq$ E2.Platform-ID (PE1 (ENROLLMENT) $\bowtie$ PE2 (ENROLLMENT)) $\bowtie$ COURSE)

Explanation

check whether the same course exist on different platform

-5 π course-Name, platform-Name, completion-Rate ($\&$ category = 'AI' $\wedge$ completion-Rate > 85 (COURSE $\bowtie$ ENROLLMENT $\bowtie$ PLATFORM))

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Explanation:-

Shows all course along with platform names where completion is more than 85

Q2

-1 Meaning:-

Display titles of unavailable books having price greater than 1000.

SQL:- SELECT title FROM Book

WHERE isAvailable = 0

AND price > 1000;

-2 Meaning:-

Shows titles of fiction book costing above 800 or all history book

SQL :- SELECT title FROM Book

WHERE (genre = 'Fiction' AND price > 800)

OR genre = 'History';

-3 Meaning:-

Show title and price of available science book

SQL :- SELECT title, price FROM Book

WHERE genre = 'Science'

AND isAvailable = 1;

-4 Meaning:- Display title and price divided by 10 of available science book

SQL:- SELECT title, price/10 FROM Book

WHERE genre = 'Science' AND isAvailable = 1;

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-5 Meaning:- Finds publisher of publisher who published the book 'AI Basics'

```
SQL:- SELECT p.Publisher-ID FROM Publisher p
      JOIN Book b ON p.bookID = b.BookID
      WHERE b.title = 'AI Basics';
```

-6 Meaning:- Show staff names and library names where staff ^{address} ~~name~~ matches library city

```
SQL:- SELECT s.name, l.libName
      FROM staff s JOIN library l
      ON s.libID = l.libID
      WHERE s.address = l.city;
```

-7 Meaning:- Get staff whose salary is not equal to 6000

```
SQL SELECT staffID FROM staff
      where salary <> 6000;
```

-8 Meaning Find publisher ID who exist both in Karachi and Islamabad

```
SQL SELECT publisherID FROM Publisher
      WHERE city = 'Karachi'
      INTERSECT
      SELECT PublisherID FROM Publishers
      WHERE city = 'Islamabad';
```

-9 Display title of book which are not published

```
SQL:- SELECT title FROM Book WHERE title NOT IN (SELECT b.title
      FROM Book b JOIN Publishers p ON p.bookID = b.bookID);
```

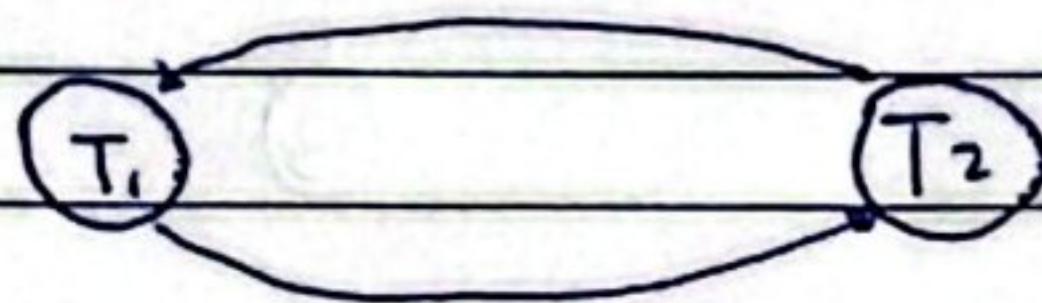
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Q3a: non-serial schedule

T_1	T_2
1 read(A)	
2	read(A)
3 $A := A - N$	
4	$A := A + M$
5 write(A)	
6	write(A)

b.



conflicts

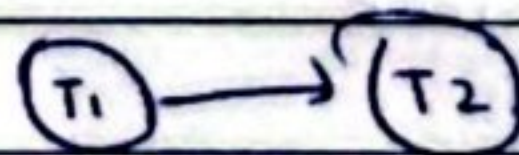
T_1 read(A) $\rightarrow$ T_2 write(A)

T_2 read(A) $\rightarrow$ T_1 write(A)

T_1 write(A) $\rightarrow$ T_1 write(A)

c. this schedule is not conflict serializable as it has a cycle ($T_1 \rightarrow T_2 \rightarrow T_1$)

d. correct schedule



T_1 T_2

1 read(A)

2 $A := A - N$

3 write(A)

4 read(A)

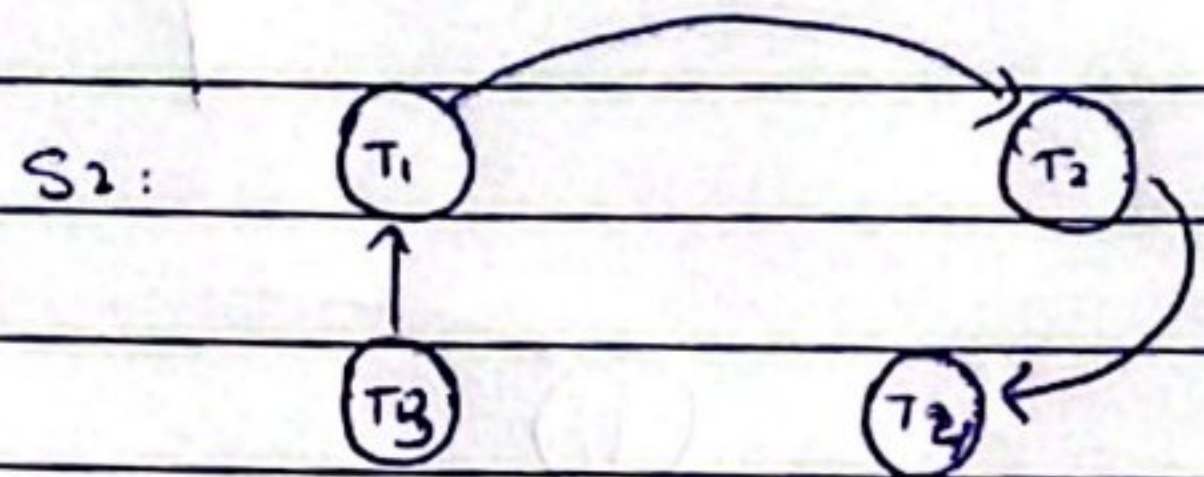
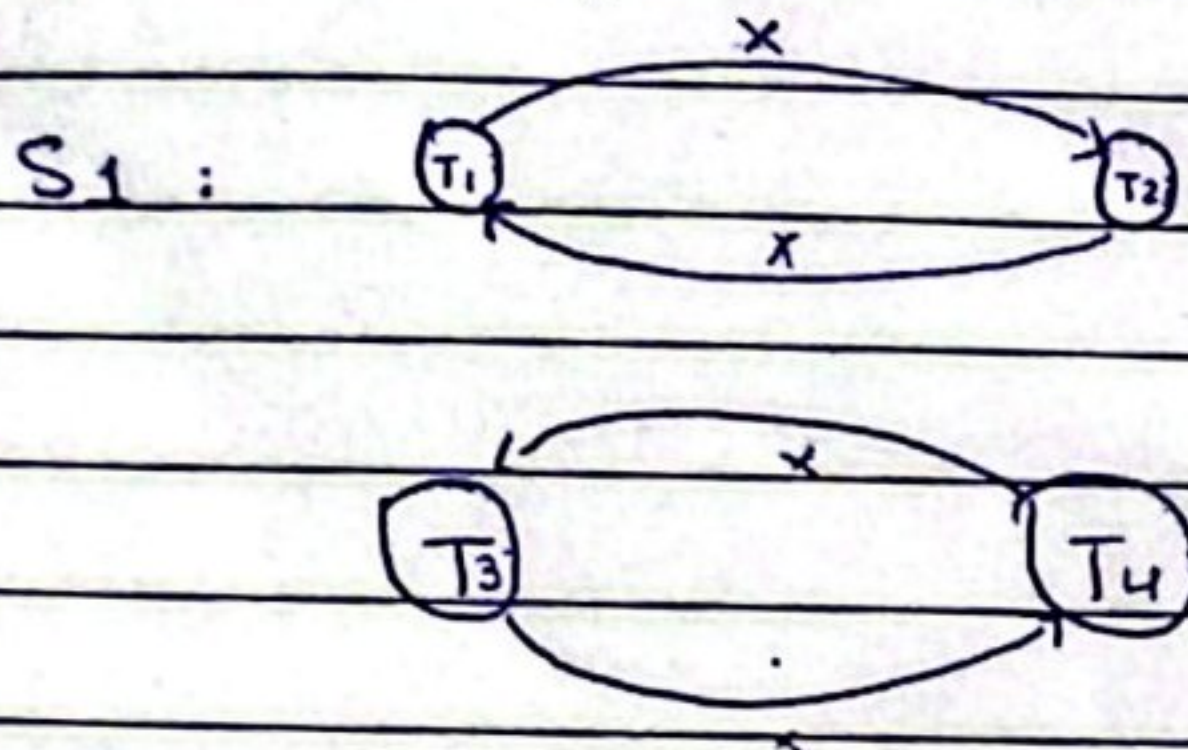
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5 $A: A + M$
6 write (A)

c. In reality, for example in banking system, two transaction occurring interleaved may cause lost update problem, when trying to access the same resource ^{and} change / write

Q4) a: precedence graph



b). S1 is not serializable
S2 is conflict serializable

c) Equivalent serial order

For S2: $T_3 \rightarrow T_1 \rightarrow T_2 \rightarrow T_4$
 $\downarrow$ starting
 from here

Q5) a- the problem is a lost update problem T_2 reads wrong value (before T_1 commits its), overwriting and losing the actual update value.

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b- T_1 reads 5000

T_2 reads 5000

$T_1 \rightarrow 5000 - 1000 = 4000$

$T_2 \rightarrow 5000 + 2000 = 7000$

T_1 write 4000

T_2 write 7000

Final balance = 7000

c- correct balance if T_2 reads correct value.

$5000 - 1000 = 4000 \rightarrow T_2$ reads 4000 correct
 $4000 + 2000 = 6000$ value.

Q6) When a transaction occurs, it is assigned a unique time stamp.

Smaller time stamp = arrived early
↓
should be executed first.

$\rightarrow$ Read-Ts(x) $\Rightarrow$ timestamp of recent transaction which reads x.

$\rightarrow$ Write-Ts(x) $\rightarrow$ timestamp of recent transaction ~~write~~ which writes x.

Rules to follow.

- if transaction T wants to read x;

if ~~only~~ $Ts(T) \geq \text{write-Ts}(x)$;

only then allow read.

and update $\text{Read-Ts}(x) = \max(\text{Read-Ts}(x), Ts(T))$;

• if transaction T wants to write x;

if $Ts(T) \geq \text{Read-Ts}(x)$ or $Ts(T) \geq \text{write-Ts}(x)$.

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Only then allow write.

and update write - $T_s(x) = T_s(T)$

Problems that can occur;

-) Cascading Restart of Transactions

-) Starvation (transaction won't be completed if newer ones conflicting).

Strict timestamp Ordering

↳ T_2 should wait till T_1 commits or aborts and should then read the value updated / corrected.